THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A ray of light strikes a plane mirror at an angle of 40° to the normal. The angle between the incident and reflected rays is: 1

(a) 50° **(b)** 40° **(c)** 100° **(d)** 80°

2. A tree casts a shadow of 10 m when a 2 m stick casts a 1 m shadow at the same time. The height of the tree is: 1

(a) 12 m **(b)** 20 m **(c)** 10 m **(d)** 5 m

3. In a pinhole camera, if the object distance is doubled, the image size becomes: 1

(a) double **(b)** half **(c)** same **(d)** quarter

4. A ray strikes the mirror surface making an angle of 25° with the surface. The angle of reflection is: 1

(a) 90° **(b)** 65° **(c)** 50° **(d)** 25°

5. The shadow of a pole is shortest at: 1

(a) night **(b)** evening **(c)** morning **(d)** noon

6. In a pinhole camera, the size of image increases when the screen is moved: 1

(a) towards the hole **(b)** away from the hole
(c) upside down **(d)** sideways

7. A ray falls along the normal. The angle of reflection is: 1

(a) 0° **(b)** 90° **(c)** 180° **(d)** 45°

8. The shadow of a pole is longest at: 1

(a) sunrise or sunset **(b)** 11 a.m.
(c) midday **(d)** noon

9. A bigger hole in a pinhole camera makes the image: 1

(a) blurred and brighter **(b)** sharper and dimmer
(c) inverted **(d)** smaller

10. A girl stands 1.5 m from a mirror. She walks 0.5 m towards it. The distance between her and her image now is: 1

(a) 1 m **(b)** 3 m **(c)** 2 m **(d)** 1.5 m

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The angle of incidence of a ray on a plane mirror is 35°. Find the angle of reflection and the angle between incident and reflected rays. 2

12. Why do we use a white screen for a pinhole camera image? 2

13. How do mirrors in a periscope make the image erect? 2

14. A girl is 2 m from a plane mirror. If she moves 1 m closer, what is the distance between her and her image? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ray of light makes an angle of 70° with the surface of a plane mirror. Find the angle of incidence, reflection and the angle between incident and reflected rays. 3

16. Why do we use a mirror at the corner of a blind road? 3

17. Explain why we see a smooth and bright image in a plane mirror but not in a wall. 3

OR

A girl 1.4 m tall stands 3 m from a mirror. State the height of her image and the distance of the image from her. 3

18. Explain why the surface of a calm lake acts as a mirror and a rough lake does not. 3

OR

Describe the use of a periscope in submarines. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A ray of light strikes a plane mirror at 60° to the surface. Draw and describe the path of the reflected ray and find all angles. 5

OR

Explain how a solar and a lunar eclipse occur using the ideas of shadow. 5

20. Explain how the size of the image in a pinhole camera changes with distance of the object and the distance between hole and screen. 5

OR

A girl walks towards a mirror at 1 m/s. At what speed does she approach her image? 5

END OF QUESTION PAPER — **Total: 40 Marks**

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